

shortest_path_length

shortest_path_length(*G*, *source=None*, *target=None*, *weight=None*) [\[source\]](#)

Compute shortest path lengths in the graph.

Parameters : **G** : NetworkX graph

source : node, optional

Starting node for path. If not specified, compute shortest path lengths using all nodes as source nodes.

target : node, optional

Ending node for path. If not specified, compute shortest path lengths using all nodes as target nodes.

weight : None or string, optional (default = None)

If None, every edge has weight/distance/cost 1. If a string, use this edge attribute as the edge weight. Any edge attribute not present defaults to 1.

Returns : length: int or dictionary

If the source and target are both specified, return the length of the shortest path from the source to the target.

If only the source is specified, return a dictionary keyed by targets whose values are the lengths of the shortest path from the source to one of the targets.

If only the target is specified, return a dictionary keyed by sources whose values are the lengths of the shortest path from one of the sources to the target.

If neither the source nor target are specified return a dictionary of dictionaries with `path[source][target]=L`, where L is the length of the shortest path from source to target.

Raises : **NetworkXNoPath**

If no path exists between source and target.

! See also

`all_pairs_shortest_path_length` , `all_pairs_dijkstra_path_length` ,

`single_source_shortest_path_length` , `single_source_dijkstra_path_length`

Notes

The length of the path is always 1 less than the number of nodes involved in the path since the length measures the number of edges followed.

For digraphs this returns the shortest directed path length. To find path lengths in the reverse direction use `G.reverse(copy=False)` first to flip the edge orientation.

Examples

```
>>> G=nx.path_graph(5)
>>> print(nx.shortest_path_length(G,source=0,target=4))
4
>>> p=nx.shortest_path_length(G,source=0) # target not specified
>>> p[4]
4
>>> p=nx.shortest_path_length(G,target=4) # source not specified
>>> p[0]
4
>>> p=nx.shortest_path_length(G) # source,target not specified
>>> p[0][4]
4
```

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